

04 · Which funnel stage to invest in? — mediation (pathmc)

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The business decision. We have a fixed budget to split across the growth funnel — **onboarding** -> **engagement** -> **activation** -> **conversion** — and want to put it where it moves paid conversion the most. The complication is that the stages form a **chain**: money spent improving onboarding doesn't only help onboarding, it *cascades* downstream into engagement, activation, and ultimately conversion. So we can't just correlate each stage with revenue; we have to trace the causal *paths* and decide where a push travels furthest.

1.0.1 The concept: mediation, direct and indirect effects

When a treatment X affects an outcome Y *through* an intermediate variable M (a **mediator**), the total effect splits in two:

- the **direct effect** — X 's influence on Y that does *not* go through M (Pearl's *natural direct effect*, NDE);
- the **indirect effect** — the part that flows $X \rightarrow M \rightarrow Y$ (the *natural indirect effect*, NIE).

Total effect = direct + indirect, and the **proportion mediated** = indirect / total tells you how much of a stage's value is really realised *downstream*. In a linear model the indirect effect is just the **product of the coefficients along the path** (the classic " $a \cdot b$ " of mediation analysis) — **pathmc** gives us exactly this via `effect(path)`, which returns the **posterior product-of-coefficients along the path** (exact for a linear SCM — not a Monte-Carlo simulation through the graph), so we get every path's effect *with uncertainty*.

1.0.2 The catch this notebook is honest about

Mediation leans on a **stronger** assumption than a simple treatment-effect analysis: no unmeasured confounding of the *mediator-outcome* link (on top of the usual treatment-outcome one). If some unmeasured trait drives both engagement *and* conversion, the direct/indirect *split* is biased even when the *total* effect is fine. So Step 6 doesn't just report the split — it **stress-tests how much such confounding would have to exist to flip the "invest here" decision**.

On real data. Swap in your **product-analytics event data** — per-user onboarding score, engagement, activation flag, conversion — which every growth team already logs. The method needs the funnel's causal *order* to be known (it usually is) and the no-hidden-mediator-confounder assumption to be defensible (Step 6 shows how to argue it).

`pathmc` builds the chain as a structural causal model and computes **path-specific effects** as the posterior **product-of-coefficients along each path** — exact for a linear SCM (we reserve “simulate through the intervened graph” for `do()/ate()`) — the modern, uncertainty-aware mediation analysis.

It follows the repo’s fixed **7-step contract**: question -> simulate -> identify -> estimate -> validate -> decide in € -> caveats.

1.1 2 · Simulate a ground truth

We simulate the funnel as a **causal chain**: `onboarding_score -> engagement -> activated -> converted`, with `channel_quality` feeding both engagement and activation (a common driver we’ll need to account for). Each arrow is a real structural equation with a planted coefficient, so we know the **true** total effect of onboarding on conversion *and how it splits across paths* — which is exactly what we’ll ask the model to recover. By construction most of onboarding’s effect flows the long way round (`engagement -> activated -> converted`), with a smaller slice going `engagement -> converted` directly. If the method can rediscover that split from data alone, we can trust it on a funnel where we *don’t* know the answer.

The data-generating model — exactly what `dgp.funnel` implements (defaults & seed in `src/cmp/dgp.py`). Onboarding $O \sim U(0,1)$ and channel quality $Q \sim U(0,1)$ are exogenous; each downstream stage is a linear structural equation:

$$\begin{aligned} \text{engagement} &= 2.2O + 1.1Q + \varepsilon_1, & \varepsilon_1 &\sim \mathcal{N}(0, 0.4^2) \\ \text{activated} &= 1.6\text{engagement} + 0.6Q + \varepsilon_2, & \varepsilon_2 &\sim \mathcal{N}(0, 0.4^2) \\ \text{converted} &= 0.9\text{activated} + 0.35\text{engagement} + \varepsilon_3, & \varepsilon_3 &\sim \mathcal{N}(0, 0.3^2). \end{aligned}$$

The planted path effects are the **products of coefficients along each route** — exactly what the mediation analysis must recover: the long path $O \rightarrow \text{eng} \rightarrow \text{act} \rightarrow \text{conv} = 2.2 \times 1.6 \times 0.9 = 3.168$, the short path $O \rightarrow \text{eng} \rightarrow \text{conv} = 2.2 \times 0.35 = 0.77$, and **no** direct $O \rightarrow \text{conv}$ arrow (true NDE = 0), so the true total effect is 3.938 and the true proportion mediated is 100%. Q feeds two stages (engagement *and* activation) — the common driver the model must include to keep the paths clean.

TRUE effects of onboarding on conversion:

<code>indirect_via_engagement_activated</code>	3.17
<code>indirect_via_engagement_only</code>	0.77
<code>total</code>	3.94

	<code>onboarding_score</code>	<code>channel_quality</code>	<code>engagement</code>	<code>activated</code>	<code>converted</code>
0	0.864798	0.726815	2.576471	3.869429	4.870321
1	0.855303	0.790067	3.206651	5.673260	6.371102
2	0.811023	0.348493	2.493785	4.691864	5.394113
3	0.261446	0.465793	0.572682	1.261547	1.264003
4	0.077199	0.887974	1.520775	3.259916	3.146544

1.2 3 · Identify — direct vs indirect effects (NDE / NIE)

For a chain $X \rightarrow M \rightarrow Y$ the **total effect** decomposes into a **natural direct effect** and **natural indirect effect** (Pearl):

$$TE = NDE + NIE, \quad NDE = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))], \quad NIE = \mathbb{E}[Y(1, M(1)) - Y(1, M(0))].$$

Under a linear SCM these are the **product-of-coefficients**: indirect = $a \cdot b$, direct = c , and `pathmc` computes each path-specific effect as exactly that product along the path (exact for the linear SCM here — not a Monte-Carlo simulation through the graph). The **proportion mediated** = NIE/TE tells you how much of onboarding’s value is “really” downstream engagement/activation. Crucially we **include a direct onboarding->converted edge in the model and test it**, so a near-100%-mediated answer is *discovered* (the coefficient comes out ~ 0), not forced by leaving the edge out.

Assumptions (stronger than a single-link analysis): no unmeasured confounding of $X-Y$, $M-Y$, and $X-M$, plus **no treatment-induced mediator-outcome confounding** — no $M-Y$ confounder is itself affected by X (the practical stand-in for the genuine *cross-world independence* assumption of Pearl/Imai). Natural effects are *not identified* if that last one fails — and the $M-Y$ confounding assumption is the one most likely to break, so we stress-test it in step 6.

1.3 4 · Estimate — fit the chain and query every path

We hand `pathmc` the **whole chain at once** — one equation per stage — rather than three disconnected regressions. That matters: because the model knows the full graph, it can propagate an intervention on onboarding *through* engagement and activation to conversion, giving genuine path-specific effects instead of naive stage-by-stage correlations. The `effects_summary()` below lists every structural coefficient (the a ’s and b ’s from Step 3); Step 5 then combines them into total, direct, and indirect effects with uncertainty.

Throughout, uncertainty is reported as a **credible interval** — the Bayesian analogue of a confidence interval, the range that contains the parameter with the stated posterior probability; note the `effects_summary()` table below shows a **94% HDI** (`hdi_3%/hdi_97%`) while Step 5’s print reports a **90%** interval.

The fitted model, in symbols — exactly the three structural equations of step 2, refit with a coefficient posterior on every arrow, *plus* one extra edge (`c_on`, a direct $O \rightarrow$ converted arrow) we include deliberately even though we suspect it is zero, so “how much is mediated?” is *discovered* (its posterior should sit at ~ 0) rather than assumed by omission. Each coefficient gets a weakly-informative normal prior, and `effect(path)` returns the **product of the posterior coefficient draws along the path** — e.g. the long path’s effect is the posterior of $e_{on} \cdot a_{eng} \cdot c_{act}$, uncertainty included.

mediation model convergence: max r-hat 1.000 - min ESS 3737 - divergences 0

	mean	sd	hdi_3%	hdi_97%
name				
e_on	2.153	0.045	2.074	2.241
e_ch	1.132	0.044	1.053	1.222
a_eng	1.617	0.018	1.585	1.652
a_ch	0.592	0.051	0.491	0.681
c_act	0.901	0.022	0.858	0.941
c_eng	0.336	0.045	0.251	0.420

c_on 0.060 0.055 -0.044 0.162

Real-data check + the naive-regression trap. Before trusting the split: (a) do the funnel graph's own testable implications survive on this data, and (b) what would a *naive* controlled regression — onboarding **plus** the mediators, all in one OLS — report for onboarding? The path queries above need neither trick; the naive regression, as the numbers show, misreads onboarding's total effect.

Testable implications of the funnel DAG: 3 checked -> 1 violated; $P(>=1 \text{ by chance at } \alpha=0.05) = 0.14$ (graph survives falsification).

the flagged independence: `channel_quality   onboarding_score | {}` (partial corr +0.087, $p = 0.006$) - in this simulation every implied independence holds by construction, so this is a knowable false positive, exactly what the Binomial budget above prices in.

Naive OLS `converted ~ onboarding + engagement + activated : onboarding coef = +0.059` (vs TRUE total 3.94).

Putting the mediators in the regression collapses onboarding's coefficient toward ~0 (it now reads only the leftover DIRECT effect, not the total). That is exactly why a naive controlled regression MISreads onboarding's real leverage -- and why the path decomposition above, not OLS, is what you need.

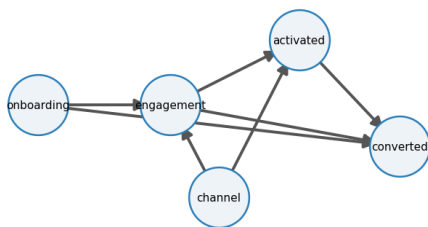
1.4 5 · Validate — recover the total, the paths, the DIRECT effect, and the proportion mediated

We query four things and check each against the planted truth: the **total** effect of onboarding on conversion; the effect flowing through each **path**; the **direct** (natural direct) effect — which the model is now free to find, and which should come out ~ 0 if onboarding really works only through the funnel; and the **proportion mediated** (indirect ÷ total). A proportion near 100% *with a direct effect whose interval includes 0* is the honest version of “onboarding works almost entirely by feeding the rest of the funnel” — a very different investment story from a direct effect.

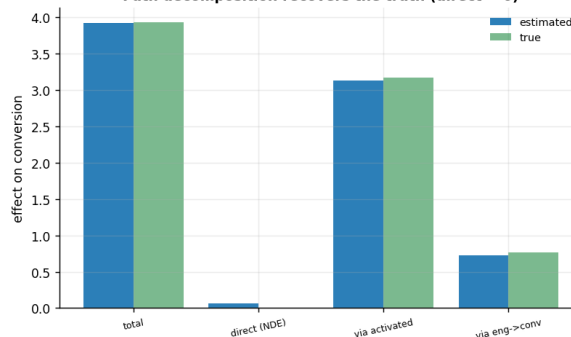
TOTAL onboarding -> converted	3.92	(true 3.94)	[90% 3.77, 4.08]
DIRECT (NDE) onboarding -> converted	0.06	(true 0.00)	[90% -0.03, 0.15]
via engagement->activated->conv	3.14	(true 3.17)	[90% 2.96, 3.31]
via engagement->converted	0.72	(true 0.77)	[90% 0.56, 0.89]

The DIRECT effect's 90% interval includes 0, so the ~98% proportion mediated is DISCOVERED - the model was free to find a direct onboarding->conversion path and the data say there isn't one.

Funnel chain (direct onboarding->converted tested, ~0)



Path decomposition recovers the truth (direct ~ 0)



How to read this. *Left* is the causal graph we fit — including a **direct** onboarding->conversion arrow, which we deliberately added so the model could find a direct effect if one existed. *Right* — the estimated total and each path effect (blue) sit on top of the planted truths (green), and the **direct (NDE) bar is ~ 0** (its interval includes zero): onboarding works almost entirely *through* the funnel, and that near-100%-mediated result is now **discovered, not assumed**. This is the decomposition a naive “regress conversion on each stage” analysis cannot give you.

1.5 6 · Decide, in euros — a *rankable* leverage metric, and its fragility

Two moves, and both correct a trap.

(1) Rank the levers — but not by “€ per one-unit change.” A one-unit move means something completely different per stage (one unit of `onboarding_score` is a large fraction of its whole range; one unit of `activated` is a fraction of a standard deviation), so a €/unit ranking is not a rankable quantity — it *reverses* when you switch to comparable **per-standard-deviation** moves. The decision-relevant metric is **ROI = (€ value of a 1-SD lift) ÷ (€ cost to produce that 1-SD lift)**; we state an illustrative equal cost per SD and report $P(\text{ROI} > 1)$. Swap in your real per-stage costs and the ranking follows.

(2) Stress-test the split — coherently. The direct/indirect decomposition rests on *no unmeasured confounding of the mediator–outcome link* (Step 3). A confounder of engagement->conversion biases the fitted edge — but that same edge carries **both** the engagement lever **and** onboarding’s total effect, so an honest sweep must shrink them *together* (a common mistake is to bias one and freeze the other). We do that per posterior draw and read off how much of the engagement->conversion edge a hidden confounder would have to explain before the leverage ranking changes.

lever	€/unit	€/SD	ROI (€/€)	P(ROI>1)
engagement	215	176	1.76	1.0
activated	108	159	1.59	1.0
onboarding	470	133	1.33	1.0

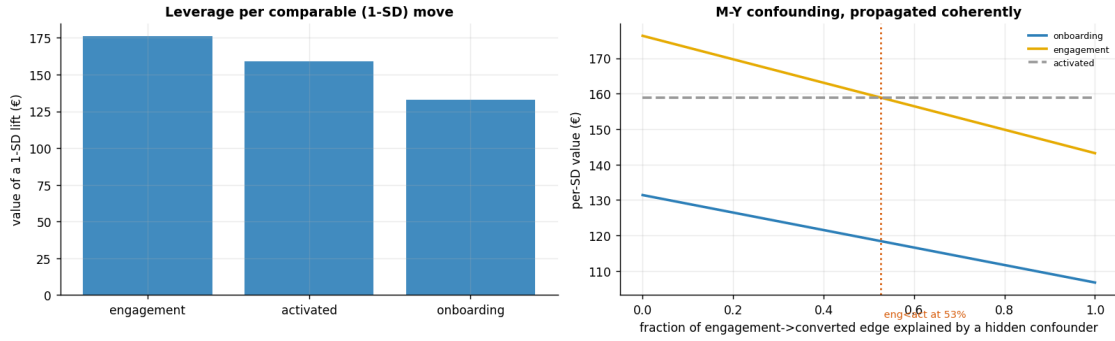
Per-UNIT vs per-SD DISAGREE: a 1-unit onboarding move is only 0.28 SD (tiny) while a

1-unit activated move is 1.47 SD (large), so €/unit is not rankable - use ROI on a stated

cost. At equal €100/SD the order is engagement > activated > onboarding and all clear break-even.

Fragility: engagement's per-SD lead over activation vanishes once a hidden M-Y confounder explains ~53% of the engagement->converted edge - a real caveat.

Onboarding (the root cause) stays robust.



1.6 7 · Caveats

- **Natural effects lean on strong assumptions** — no unmeasured confounding of any of the three links plus the cross-world condition. Step 6 quantified *coherently* (shrinking both affected levers together) how much $M-Y$ confounding would flip the leverage ranking; the point estimate alone hides that the engagement-vs-activation ordering is the fragile one.
- **A €/unit leverage ranking is not a rankable quantity** when the stage metrics live on different scales — it reverses under per-SD units. Decide on **ROI against a stated cost-to-move**, not €/unit.
- **Don't control away the mechanism.** The goal is to *decompose* the path; adjusting for a mediator when you wanted the total effect removes the very thing you're pricing (nb 05).
- **Linear-SCM decomposition is exact only if the SCM is linear.** For strong nonlinearities lean on `pathmc`'s numbers, not hand-multiplied coefficients.